

Yuxiang (Kevin) Zheng

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ABOUT ME

I am a PhD candidate at the University of Technology Sydney, with a strong background in machine learning theory, mathematical modelling, and data-driven algorithm design. My research interests mid/disinformation detection, robust learning, and generalisation in real-world systems. I have contributed as a reviewer for top-tier conferences such as ICLR, ICML, CVPR, and AAAI, and I am committed to advancing both the theoretical foundations and practical applications of machine learning.

EDUCATION AND TRAINING

Doctor of Philosophy

University of Technology Sydney [01/09/2025 – Current]

City: Sydney | Country: Australia

The University of Sydney

Bachelor of Advanced Computing (Honours) [01/03/2021 – 06/2025]

City: Sydney | Country: Australia | Field(s) of study: Computational Data Science

Dalyell Scholar, recognized for outstanding academic performance.

The University of Sydney

Bachelor of Science [01/03/2021 – 06/2025]

City: Sydney | Country: Australia | Field(s) of study: Mathematics

Swiss Federal Institute of Technology Zurich (ETHz)

Worldwide Exchange Programme [18/09/2023 – 03/03/2024]

City: Zurich | Country: Switzerland | Field(s) of study: Mathematics ; Machine Learning Theory

WORK EXPERIENCE

🏢 The University of Sydney – Sydney, Australia

City: Sydney | Country: Australia

Casual Academic

[06/2024 – Current]

- Delivered lectures and tutorials for **COMP5328: Advanced Machine Learning**, emphasizing **mathematical foundations** and **theoretical concepts** such as **convex optimization**, **statistical learning theory**, and **generalization bounds**.
- Guided students in deriving and implementing machine learning algorithms, bridging theoretical insights with practical applications using **Python** and **PyTorch**.
- Designed and assessed assignments to deepen students' understanding of **probabilistic models**, **optimization techniques**, and **high-dimensional data analysis**.
- Delivered a **guest lecture** on **Large Language Models (LLMs)** and **GPT architecture**, covering advancements in **transformers** and addressing industrial challenges like **scalability** and **bias mitigation**.

🏢 Trustworthy Machine Learning Lab – Sydney, Australia

City: Sydney | Country: Australia

Research Assistant

[05/2022 – 06/2025]

- Led independent research projects in machine learning, focusing on **noisy label** and **semi-supervised learning**, addressing key challenges in data reliability.
- Designed and implemented machine learning algorithms using **PyTorch**, achieving measurable improvements in experimental results.
- Conducted entire research workflows, including **experiment design, data preprocessing, model training, and result analysis**.
- Served as a **peer reviewer** for top-tier AI/ML conferences, including **ICLR, ICML, CVPR, ACMMM, and AAAI**, ensuring the quality and rigour of cutting-edge research.

PROJECTS

[25/10/2023 – 2024]

Noisy label Learning Project

- **Spearheaded the entire project lifecycle**, from conceptualizing the research problem to developing and validating a novel framework for **noisy label learning**.
- Independently proposed and **mathematically validated** a novel framework leveraging **generative AI** to address noisy label challenges, introducing a **unique sample filtering method** to enhance data quality, learning efficiency, and model robustness.
- Designed and executed comprehensive experiments, ensuring scalability and robustness, resulting in a **7% improvement over state-of-the-art benchmarks**.
- Delivered all aspects of the project, including **problem formulation, algorithm implementation (PyTorch)**, and **experimental analysis**, demonstrating end-to-end research capability.

Link: <https://github.com/KevinCarpricorn/NeurIPS2024>

[01/12/2022 – 09/03/2023]

Bayes Transition Partial-label Learning Project

- **Proposed a Bayesian framework** to address **partial-label learning** by modelling **transition probabilities** between observed and true labels.
- Conducted **theoretical validation** and implemented experiments using **PyTorch**, improving data-label reliability under semi-supervision.
- Gained expertise in **probabilistic modelling** and **Bayesian inference**, bridging mathematical theory with practical machine-learning applications.

Link: https://github.com/KevinCarpricorn/Transition_Matrix_PLL

[01/08/2022 – 26/11/2022]

Partial-label Continual Learning Project

- **Developed a continual learning algorithm** for partial-label scenarios, integrating **memory-based techniques** to enhance task retention and minimize catastrophic forgetting.
- Validated the algorithm across multiple datasets, achieving **comparable performance** to fully supervised methods in dynamic environments.
- Demonstrated expertise in **multi-task optimization** and **real-time model adaptation** for evolving tasks.

Link: <https://github.com/KevinCarpricorn/PLCL>

HONORS & CERTIFICATIONS

[08/2025]

University Medal

Awarded the University Medal by the University of Sydney in recognition of outstanding academic achievement in the Bachelor of Advanced Computing (Honours), Computational Data Science. This is the University's highest undergraduate honour, granted to graduates who have demonstrated exceptional academic performance and research excellence.

[01/2022]

Dalyell Scholar

The Dalyell Scholars program is a prestigious initiative for high-achieving students at the University of Sydney. It offers a range of advanced learning opportunities and global experiences. As a recipient of this honour, I was recognized for my academic excellence and potential for leadership, both of which are crucial for the data analyst role I am applying for.

[01/2025 – 01/2028]

AWS Certified Machine Learning – Specialty

Earners of this certification have an in-depth understanding of AWS machine learning (ML) services. They demonstrated ability to build, train, tune, and deploy ML models using the AWS Cloud. Badge owners can derive insight from AWS ML services using either pretrained models or custom models built from open-source frameworks.

Link: https://www.credly.com/badges/b324db7a-0cb4-41b9-b3a7-a3275dde8f6b/public_url

SKILLS

Programming Languages

Python / R / SQL

Machine Learning Frameworks & Libraries

Pytorch / Numpy / Pandas / Matplotlib

Platforms and Tools

AWS Cloud / Git / Jupyter Notebook/Jupyter Lab / Latex / Linux